

New permanent SCON attribute for COMP364: Artificial Intelligence

upper-level CS elective - requires 3 prev. courses & programming/algorithms

John MacGinnick

SCON Learning outcome: "Think critically about a sustainability qu./prob./soln."

Sustainability content: new topic "AI + Sustainability" (2x 75min class meetings + 1 assignment)

Required watching: Dson Clarke Forum panel, "Real-World effects of AI" 2/26/2026

Required reading: excerpt of Johnson (2024) What if we get it right?
- chapter "The AI deluge" - interview with Mustafa Suleyman

Optional field trip: previous weekend, do a drive-by of the PAX-1 site →

Class 1: a) Socratic discussion: "What is sustainability?"

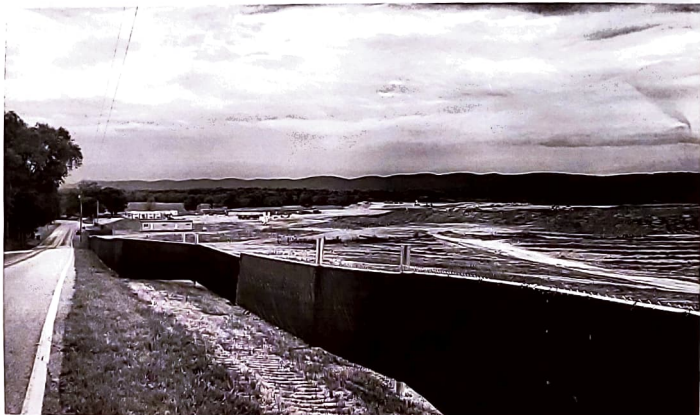
b) POGIL-style think-group-share:

- 3 topics, reports for each group
- 1. What are syst. impacts of large data centers?
- 2. ——— trying a hyperscale LM?
- 3. ——— a consumer using A.I.

What could an alternate experience look like for the drive-by of the PAX-1 site? It could be so! It could be for mt. power for mt. was. you. not.

Week	Day	Date	topic	Class number	home work due
1	Tue	9/1		class 1	
	Fri	9/4		class 2	
2	Tue	9/8	1. Classical search	class 3	
	Fri	9/11		class 4	
3	Tue	9/15	2. Elementary classifiers: nearest neighbor and decision tree	class 5	HW1
	Fri	9/18	ThinkingAI1-4 + Turing1950	class 6	
4	Tue	9/22		class 7	HW2
	Fri	9/25	3. Neural networks and backpropagation	class 8	
5	Tue	9/29		class 9	
	Fri	10/2		class 10	HW3
6	Tue	10/6	4. Convolutional neural networks	class 11	
	Fri	10/9	exam review	class 12	HW4
7	Tue	10/13	midterm exam 1	class 13	
	Fri	10/16	ThinkingAI5-7	class 14	
	Tue	10/20	fall break		
8	Fri	10/23		class 15	
	Tue	10/27	5. Reinforcement learning	class 16	
9	Fri	10/30		class 17	
	Tue	11/3	6. AI and sustainability	class 18	HW5
10	Fri	11/6	ThinkingAI8-10	class 19	
	Tue	11/10		class 20	HW6
11	Fri	11/13	7. Transformers and large language models	class 21	
	Tue	11/17		class 22	
12	Fri	11/20	ThinkingAI11-14 + searle1980	class 23	
	Tue	11/24	exam review	class 24	HW7
	Fri	11/27	thanksgiving vacation		
13	Tue	12/1	midterm exam 2	class 25	
	Fri	12/4		class 26	
14	Tue	12/8	8. Final project	class 27	
	Fri	12/11		class 28	
		12/18, Fri 2pm-5pm	final presentations		

AMAZING!
WILL YOU ADDRESS THE ENVIRONMENTAL JUSTICE ISSUES SURROUNDING A.I. DATA CENTERS?



- c) choose asst option - group into teams.
- ① Investigative journalism - PAX-1
- ② Decide where in PA to build a new data center
- ③ Any application of AI to sustainability

future work:
- expand to 3 class meetings
- CUT some other content.

course uses AI book → (for discussions, esp. philosophy topics)

Class 2: low-tech presentation (whiteboard only... or maybe paper posters?)